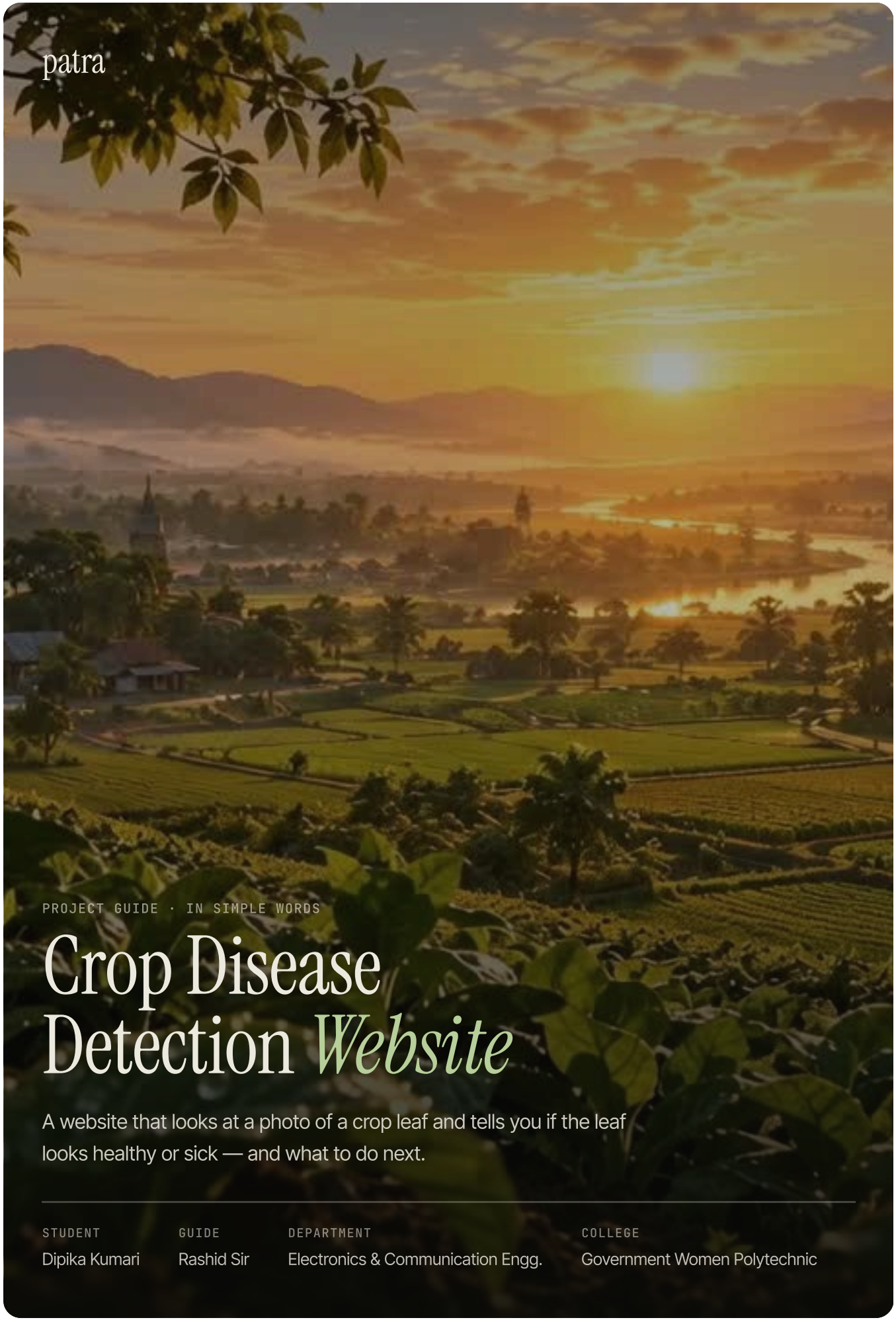


patra

PROJECT GUIDE · IN SIMPLE WORDS

Crop Disease Detection *Website*

A website that looks at a photo of a crop leaf and tells you if the leaf looks healthy or sick — and what to do next.

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1 · THE IDEA

What is this *project?*

Crops get diseases from fungus, bacteria and poor nutrition. These diseases first show up as **spots, yellow patches or white powder on the leaves**. If a farmer catches it early, the crop can be saved. If it is caught late, the disease spreads across the whole field.

Most farmers check leaves only by eye, and an expert is not always nearby. So we built **Patra** (पत्र means "leaf") — a website where you upload a leaf photo and, in less than a second, it tells you:

- Is the leaf **healthy** or does it look **diseased**?
- Which disease is **most likely**, and how sure the system is (confidence %)
- How much of the leaf is damaged (severity)
- What the farmer should do next

Important: Patra gives a **first opinion**, like a quick check-up. It does not replace an agriculture expert. Serious cases should still be shown to the local agriculture officer.

2 · THE WEBSITE

What's *inside* the website

The website has 6 pages. You move between them using the menu at the top.

PAGE	WHAT IT DOES
Home	A video that plays as you scroll — it starts with a wide farm view and zooms into one sick leaf. Then it explains the problem and the idea.
Detect	The main tool. Upload, drag-and-drop or take a leaf photo (or try a sample). You get the result with a coloured "damage map".
Disease library	8 conditions (Early Blight, Late Blight, Leaf Spot, Rust, Powdery Mildew, Leaf Blast, Nutrient Chlorosis, Healthy) with symptoms, causes and treatment. You can search and filter.
How it works	Shows the system design, the steps, hardware and software — taken from the project report.
Project	Report summary, objectives, limitations, future scope and a button to download the full report.
Scan log	Every leaf you scan is saved here with its result and time, so you can track a field over days.

3 · THE BRAIN

How the leaf checker *works*

When you upload a photo, a Python program reads it like a set of numbers (every pixel has a colour value) and follows these steps:

1. **Photo check —** Is the photo blurry? Too dark? Too bright? If yes, it shows a warning instead of giving a confident wrong answer.
2. **Colour change —** The photo is changed from RGB colours to HSV colours. HSV makes it easy to ask "is this pixel green, yellow or brown?"
3. **Find the leaf —** Green pixels are grouped together to find the leaf and separate it from soil, sky and background. If no leaf is found, it says "No leaf detected".
4. **Find the damage —** Inside the leaf, it marks brown spots, orange rust, yellow patches, white powder and very dark dead tissue.
5. **Count the spots —** It counts how many spots there are and how big they are. Many small spots and a few big spots point to different diseases.
6. **Decide —** All these measurements are turned into a score for each of the 7 classes. The highest score wins and is shown as a percentage.



What you upload



What the program sees → Early Blight, 86%

Brown spot Rust Dead tissue Yellowing Powder

Is this AI / ML? Yes — it is **computer vision** (teaching a computer to understand images). Right now the rules ("brown + few big spots = Early Blight") were set by us after testing. The next step is a **CNN (deep learning model)** that learns these rules by itself from thousands of labelled leaf photos, for example the free PlantVillage dataset. The website is already built so the CNN can be plugged in later.

4 · THE SCROLLING VIDEO

How the video plays when you *scroll*



1. **Make the video —** A 10-second farm video was created with AI (Google Gemini).
2. **Cut it into photos —** The video was cut into 250 still pictures (frames), like a flip-book.
3. **Remove the Gemini logo —** Every frame had a small Gemini star in the corner. A Python + OpenCV program found the star, worked out how see-through it was, and removed it from all 250 frames.
4. **Play on scroll —** JavaScript watches how far down you have scrolled and shows the matching frame. Scroll down = video plays forward. Scroll up = it plays backward.

5 · WHAT WE USED

Tools & *technology*

TOOL	WHAT IT DOES IN THIS PROJECT
Python `language`	The main language. The leaf checker and the server are written in Python.
Flask	A small Python web framework. It runs the website and receives the uploaded photos.
NumPy	Does the maths on the pixel values (the image is just a big table of numbers).
Pillow (PIL)	Opens, resizes and colour-converts the photos.
OpenCV	Computer-vision library used to clean the video frames (logo removal).
SQLite	A small database file that stores the Scan log.
HTML, CSS, JavaScript	Build the pages, the design, the scrolling video and the upload screen.
Google Gemini `AI`	Generated the farm-to-leaf video used on the home page.
Claude Code `AI`	An AI coding assistant by Anthropic (model: Claude Opus 5.5). It planned the website from the project report, wrote the code, designed the pages, built the leaf checker, removed the logo and tested everything.

Did it *pass the test?*

We tested the leaf checker on **24 cases** and all 24 passed: healthy and sick leaves from the video, leaves edited to look like each disease, blurry and dark photos, non-leaf photos (sky, wall, noise) and broken files. Each scan takes about **0.14 seconds**.

Limits (be honest in the viva)

- Works best with **one leaf, in daylight, filling the photo**.
- Diseases that look alike (Leaf Spot vs Early Blight) can get mixed up.
- It is not yet tested on a big set of real field photos.

What can be added next

- A trained CNN model (TensorFlow / PyTorch)
- Mobile app that works without internet
- Hindi and voice guidance for farmers
- Weather and soil sensors, GPS disease map

How to run the website

Open the project folder in a terminal and type:

```
pip install -r requirements.txt
python app.py
```

Then open `http://localhost:5000` in the browser.

Quick viva answers

Q. Why HSV and not RGB?

In HSV, the colour (hue) is separate from brightness, so "green" stays green even in shade or sunlight.

Q. How does it know it's not a leaf?

If less than 8% of the photo is leaf-coloured, it says "No leaf detected" instead of guessing.

Q. What is the confidence %?

Each disease gets a score; the scores are turned into percentages that add up to 100 (softmax). The top one is the answer.

Q. How is it different from a CNN?

Here we chose the features (colour, spot count, spot size). A CNN finds its own features by training on thousands of photos.